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Light-Based Processors

Technology, trade-offs & engineering roles

Yanjie Zhou · Siheng Ye · Jiaqi Luo · Haoran Li

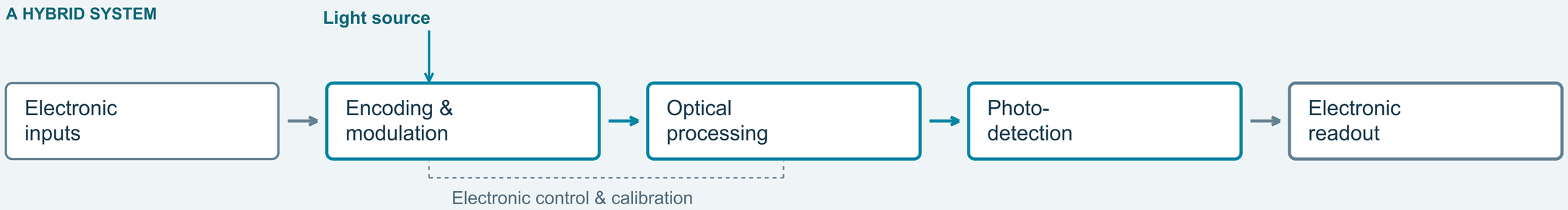
University of Nottingham Ningbo China · Department of Electrical and Electronic Engineering

01 Computing with light

Photonic circuits can perform selected linear operations, including matrix transformations, by encoding and controlling light. A practical system also needs electronic interfaces, control and readout. Different architectures place computation in different parts of this chain. [1, 2]

02 Connecting with light

Optical I/O moves information between electronic processors. Intel's 2024 optical I/O chiplet demonstration is an interconnect example: its data-transfer result should be distinguished from arithmetic performed by a photonic processor. [3]



Illustrative hybrid photonic-computing chain; architectures vary. Original explanatory schematic for this public edition, informed by [1, 2].

System view. Assess the whole system: optical loss, noise and precision, calibration, lasers, electronic conversion and data movement all matter. [1, 2]

FOUR ENGINEERING PERSPECTIVES

01

Device development

How does the device behave?

Characterize waveguides, modulators and detectors; understand optical loss, thermal sensitivity and fabrication constraints.

02

Product management

Which problem is worth solving?

Define a suitable workload, required precision and integration needs; assess useful system-level benefit.

03

Cost engineering

What makes a system viable?

Consider fabrication, packaging, light sources, electronics, yield, test and production volume together.

04

Applications engineering

How can the system be used?

Connect the optical subsystem to electronic and software interfaces; support calibration and representative benchmarks.

SELECTED PRIMARY SOURCES

[1] Y. Shen et al. Deep learning with coherent nanophotonic circuits. Nature Photonics 11, 441–446 (2017).
DOI: 10.1038/nphoton.2017.93

[2] R. Hamerly, L. Bernstein, A. Sludds, M. Soljačić and D. Englund Large-Scale Optical Neural Networks Based on Photoelectric Multiplication. Physical Review X 9, 021032 (2019).
DOI: 10.1103/PhysRevX.9.021032

[3] Intel Corporation Intel Demonstrates First Fully Integrated Optical I/O Chiplet. 26 June 2024.
Intel Newsroom · source article